

Allard Regularity Theorem

Yiming Chen

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1 Introduction

We shall prove the following theorem:

Theorem 1.1. *Let $V = \underline{v}(M, \theta)$ is a rectifiable n -varifold. If the generalized mean curvature of V is in $L^p_{loc}(\mu_V)$ in U for some $p > n$, and if $\theta \geq 1$ μ_V -a.e. in U , if $\xi \in \text{spt}V \cap U$, and if $\omega_n^{-1} \rho^{-n} \mu_V(B_\rho(\xi))$ is sufficiently close to 1 for some sufficiently small ρ , then V is regular near ξ in the sense that $\text{spt}V$ is a $C^{1,1-\frac{n}{p}}$ n -dimensional embedded submanifold near ξ .*

2 Rectifiable n -varifolds

2.1 Countably n -Rectifiable Sets

Definition 2.1. A set $M \subset \mathbb{R}^{n+k}$ is said to be countably n -rectifiable, if

$$M \subset M_0 \cup \left(\bigcup_{j=1}^{\infty} F_j(\mathbb{R}^n) \right),$$

where $\mathcal{H}^n(M_0) = 0$ and $F_j : \mathbb{R}^n \rightarrow \mathbb{R}^{n+k}$ are Lipschitz functions for any $j = 1, 2, \dots$.

Countably n -rectifiable sets have tangent spaces.

Definition 2.2. If M is an $\mathcal{H}^n$ -measurable subset of $\mathbb{R}^{n+k}$ with $\mathcal{H}^n(M \cap K) < \infty$ for any compact subset $K \subset U$, then we say that an n -dimensional subspace of P of $\mathbb{R}^{n+k}$ is the approximate tangent space for M at $x \in \mathbb{R}^{n+k}$ if

$$\lim_{\lambda \rightarrow 0^+} \int_{\eta_{x,\lambda}(M)} f(y) d\mathcal{H}^n(y) = \int_P f(y) d\mathcal{H}^n(y), \quad \forall f \in C_c^0(\mathbb{R}^{n+k}),$$

where

$$\eta_{x,\lambda} : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^{n+k}, \quad \eta_{x,\lambda}(y) = \lambda^{-1}(y - x), \quad \forall x, y \in \mathbb{R}^{n+k}, \quad \lambda > 0.$$

Below is an important characterization of countably n -rectifiable sets, but we will not prove it here.

Theorem 2.3. *Suppose M is $\mathcal{H}^n$ -measurable with $\mathcal{H}^n(M \cap K) < \infty$ for each compact $K \subset \mathbb{R}^{n+k}$. Then M is countably n -rectifiable $\iff$ the approximate tangent space $T_x M$ exists for $\mathcal{H}^n$ -a.e. $x \in M$.*

2.2 Rectifiable n -varifolds

Definition 2.4. *Let $U \subset \mathbb{R}^{n+k}$ be an open set and $M \subset U$ a countably n -rectifiable, $\mathcal{H}^n$ -measurable subset of $\mathbb{R}^{n+k}$, and let θ be a positive $\mathcal{H}^n$ -measurable function on M with*

$$\int_{M \cap K} \theta d\mathcal{H}^n < \infty$$

for each compact set $K \subset U$. For any such a pair (M, θ) , we define the rectifiable n -varifold $\underline{v}(M, \theta)$ to be the equivalence class of all pairs $(\tilde{M}, \tilde{\theta})$, where $\tilde{M} \subset U$ is countably n -rectifiable, $\mathcal{H}^n(M \Delta \tilde{M}) = 0$, and $\theta = \tilde{\theta}$ $\mathcal{H}^n$ -a.e. on $M \cap \tilde{M}$.

$V = \underline{v}(M, \theta)$ is referred to as a rectifiable n -varifold in U , and θ is called the multiplicity function of V .

$V = \underline{v}(M, \theta)$ is called an integer multiplicity rectifiable varifold, if its multiplicity function is positive integer-valued $\mathcal{H}^n$ -a.e.

There is a Radon measure μ (called the weight measure of V) on U defined by

$$\mu_V = \mathcal{H}^n \llcorner \theta,$$

and for any $\mathcal{H}^n$ -measurable set $A \subset U$,

$$\mu_V(A) = \int_{A \cap M} \theta d\mathcal{H}^n.$$

We call the mass (or weight) of the varifold V , $\mathbb{M}(V)$ defined by

$$\mathbb{M}(V) = \mu_V(U) = \int_M \theta d\mathcal{H}^n.$$

2.3 Variation of Rectifiable n -varifolds

For any rectifiable n -varifold $V = \underline{v}(M, \theta)$ in an open set $U \subset \mathbb{R}^{n+k}$, assume $\phi : (-\epsilon, \epsilon) \times U \rightarrow \mathbb{R}^{n+k}$ for some $\epsilon > 0$ is a C^1 map such that

$$\begin{cases} \phi(0, x) = x \text{ for each } x \in U, \text{ and there exists a compact set } K \subset U \text{ with} \\ \phi(t, x) = x \text{ for all } (t, x) \in (-\epsilon, \epsilon) \times (U - K), \end{cases}$$

and such that the velocity $\frac{\partial \phi(t, x)}{\partial t}$ is also C^1 . Then the initial velocity $X = \frac{\partial \phi(t, x)}{\partial t}|_{t=0}$ is a C^1 vector field with compact support in U . On the other hand, given any C^1 vector field X with compact support U , we can construct $\phi(t, x) = x + tX(x)$ and $\epsilon > 0$ sufficiently small.

Now compute the time-dependent area

$$\mathbb{M}(\phi_{t\#}(V \lrcorner K)) = \int_{M \cap K} J_M \phi_t \theta d\mathcal{H}^n.$$

Locally, for any $y \in K$, take $\tau_1, \dots, \tau_n$ the local unit orthogonal frame, we have

$$\phi_t(y) = y + t(X(y) + E(t, y)),$$

where

$$\sup_{y \in K} \left(|E(t, y)| + \sup_{\tau \in T_y M, |\tau|=1} |D_\tau E(t, y)| \right) \rightarrow 0$$

as $t \rightarrow 0$. So we have

$$D_{\tau_i} \phi(y) = \tau_i + t(D_{\tau_i} X(y) + E_i(t, y)),$$

where $\sup_{y \in K} |E_i(t, y)| \rightarrow 0$ as $t \rightarrow 0$. Hence

$$J\phi_t = \sqrt{\det(\delta_{ij} + t(\tau_i \cdot D_{\tau_j} X(y) + \tau_j \cdot D_{\tau_i} X(y) + E_{ij}(t, y)))},$$

where $\sup_{y \in K} |E_{ij}(t, y)| \rightarrow 0$ as $t \rightarrow 0$. Using the formula

$$\det(I + tA) = 1 + t \operatorname{trace} A + O(t^2)$$

as $t \rightarrow 0$, we have

$$\begin{aligned} & \det(\delta_{ij} + t(\tau_i \cdot D_{\tau_j} X(y) + \tau_j \cdot D_{\tau_i} X(y) + E_{ij}(t, y))) \\ &= 1 + t \operatorname{trace}(\tau_i \cdot D_{\tau_j} X(y) + \tau_j \cdot D_{\tau_i} X(y) + E_{ij}(t, y)) + O(t^2) \\ &= 1 + 2t \sum_{i=1}^n \tau_i \cdot D_{\tau_i} X(y) + t \operatorname{trace} E_{ij}(t, y) + O(t^2) \end{aligned}$$

as $t \rightarrow 0$.

Therefore, we can compute the first variation:

$$\begin{aligned} \frac{d}{dt} \Big|_{t=0} \mathbb{M}(\phi_{t\#}(V \lrcorner K)) &= \frac{d}{dt} \Big|_{t=0} \int_{M \cap K} J_M \phi_t \theta d\mathcal{H}^n \\ &= \frac{d}{dt} \Big|_{t=0} \int_{M \cap K} \sqrt{\det(d\phi_t)^* \circ (d\phi_t)} \theta d\mathcal{H}^n \\ &= \int_{M \cap K} \operatorname{div}_M X \cdot \theta d\mathcal{H}^n \\ &= \int_{M \cap K} \operatorname{div}_M X d\mu_V, \end{aligned}$$

where

$$\operatorname{div}_M X = \sum_{j=1}^{n+k} e_j \cdot (\nabla^M X^j).$$

Definition 2.5. We say that $V = \underline{v}(M, \theta)$ is stationary in U if

$$\frac{d}{dt}\bigg|_{t=0} \mathbb{M}(\phi_{t\#}(V \llcorner K)) = 0$$

for every family $\phi(t)$ satisfying the conditions above; or equivalently, if

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

for any C^1 vector field X having compact support in U .

Definition 2.6. Let $V = \underline{v}(M, \theta)$ be a rectifiable n -varifold in the open set $U \subset \mathbb{R}^{n+k}$. Then we say that V has generalized mean curvature vector $\underline{H}$ in U if

$$\int_M \operatorname{div}_M X \, d\mu_V = - \int_M X \cdot \underline{H} \, d\mu_V, \quad \forall X \in C_c^1(U, \mathbb{R}^{n+k}).$$

Obviously, V is stationary in U if and only if it has generalized mean curvature zero.

2.4 Monotonicity Formulae

2.4.1 Stationary Case

For a stationary rectifiable n -varifold $V = \underline{v}(M, \theta)$ in U , i.e. its generalized mean curvature $\underline{H} = 0$, we have

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

whenever X is a C^1 vector field on U with compact support in U . Fix a point $\xi \in U$, choose test vector field $X = \gamma(r)(x - \xi)$, $r = |x - \xi|$, and $\gamma : \mathbb{R}^n \rightarrow [0, 1]$ is a C^1 function with

$$\gamma'(t) \leq 0; \quad \gamma(t) = 1 \text{ when } t \leq \rho; \quad \gamma(t) = 0 \text{ when } t \geq R,$$

where $R > \rho > 0$ and $B_R(\xi) \subset U$.

For any $f \in C^1(U)$ and any $x \in M$ such that $T_x M$ exists, we have

$$\nabla^M f(x) = \sum_{j,l=1}^{n+k} e^{jl} \partial_l f(x) e_j,$$

where $\{e^{jl}\}_{j,l=1}^{n+k}$ is the matrix of the projection onto $T_x M$. We have

$$\operatorname{div}_M X = \sum_{j=1}^{n+k} \nabla_j^M X^j = \gamma(r) \sum_{j=1}^{n+k} e^{jj} + r\gamma'(r) \sum_{j,l=1}^{n+k} e^{jl} \cdot \frac{x^j - \xi^j}{r} \cdot \frac{x^l - \xi^l}{r}.$$

Since $\text{trace}\{e^{jl}\} = n$, and

$$\sum_{j,l=1}^{n+k} e^{jl} \cdot \frac{x^j - \xi^j}{r} \cdot \frac{x^l - \xi^l}{r} = \left| p_{T_x V} \left(\frac{x - \xi}{r} \right) \right|^2,$$

(Noting that $p_{T_x V} \circ p_{T_x V} = p_{T_x V}$), then we get

$$n \int_M \gamma(r) d\mu + \int_M r \gamma'(r) |\nabla^M r|^2 = 0.$$

Take $\epsilon \in (0, 1)$ and a C^1 function $\phi : \mathbb{R} \rightarrow [0, 1]$ such that $\phi(t) \leq 1$ when $t \leq 1$; $\phi(t) = 0$ when $t \geq 1 + \epsilon$, and $\phi'(t) \leq 0$. Then we take $\gamma(r) = \phi(\frac{r}{\rho})$ in the above identity provided $(1 + \epsilon)\rho < R$. Since

$$r \gamma'(r) = \frac{r}{\rho} \phi'(\frac{r}{\rho}) = -\rho \partial_\rho \phi(\frac{r}{\rho}),$$

we have

$$n \int_M \phi(\frac{r}{\rho}) d\mu - \rho \partial_\rho \int_M |\nabla^M r|^2 \phi(\frac{r}{\rho}) d\mu.$$

On the other hand, $p_{T_x V} v = v - p_{(T_x V)^\perp} v$ for any $v \in \mathbb{R}^{n+k}$, so

$$\left| p_{T_x V} \left(\frac{x - \xi}{r} \right) \right|^2 = 1 - |D^\perp r|^2,$$

where $D^\perp r$ denotes the orthogonal projection of $\frac{x - \xi}{r} = Dr$ onto $(T_x V)^\perp$. So

$$nI(\rho) - \rho I'(\rho) = -\rho \int_M \partial_\rho \left[\phi(\frac{r}{\rho}) \right] |D^\perp r|^2 d\mu,$$

where $I(\rho) = \int_M \phi(\frac{r}{\rho}) d\mu$. Multiplying ρ^{-n-1} on the two sides, then

$$\frac{d}{d\rho} (\rho^{-n} I(\rho)) = \int_M \rho^{-n} \partial_\rho \left[\phi(\frac{r}{\rho}) \right] |D^\perp r|^2 d\mu.$$

Since $(1 + \epsilon)^{-n} J'(\rho) \leq \frac{d}{d\rho} (\rho^{-n} I(\rho)) \leq J'(\rho)$, $J(\rho) = \int_M r^{-n} |D^\perp r|^2 \phi(\frac{r}{\rho}) d\mu$, integrating over $[\sigma, \rho]$, we obtain

$$(1 + \epsilon)^{-n} (J(\rho) - J(\sigma)) \leq \rho^{-n} I(\rho) - \sigma^{-n} I(\sigma) \leq J(\rho) - J(\sigma).$$

Now let $\epsilon \rightarrow 0$, then $I(\rho) \rightarrow \int_{0 \leq r \leq \rho} d\mu = \mu(B_\rho(\xi))$, and $J(\rho) \rightarrow \int_{B_\rho(\xi)} r^{-n} |D^\perp r|^2 d\mu$. Then we get the monotonicity formula

$$\frac{1}{\omega_n} \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} |D^\perp r|^2 = \frac{\mu(B_\rho(\xi))}{\omega_n \rho^n} - \frac{\mu(B_\sigma(\xi))}{\omega_n \sigma^n}.$$

Particularly, $\frac{1}{\omega_n \rho^n} \mu(B_\rho(\xi))$ is increasing in ρ for all $0 < \sigma \leq \rho < R$ whenever $B_R(\xi) \subset U$.

2.4.2 Bounded Generalized Mean Curvature

Now we want to obtain more results. If we take the test vector field $X = h(x)\gamma(r)(x - \xi)$ where h is nonnegative C^1 function, then in fact we get

$$\int_M \left(n\gamma(r) + r\gamma'(r) |\nabla^M r|^2 \right) h d\mu = - \int_M (x - \xi) \cdot (h\underline{H} + \nabla^M h) \gamma(r) d\mu.$$

Replacing $\gamma(r)$ by $\phi(\frac{r}{\rho})$ similarly, we get

$$n \int_M h \phi \left(\frac{r}{\rho} \right) d\mu - \rho \frac{d}{d\rho} \int_M h |\nabla^M r|^2 \phi \left(\frac{r}{\rho} \right) d\mu = E_h(\rho),$$

where $E_h(\rho) = \rho^{-n-1} \int_M (h\underline{H} + \nabla^M h) \cdot (x - \xi) \phi \left(\frac{r}{\rho} \right) d\mu$.

Now suppose that there exists a constant $\Lambda > 0$ such that

$$R \sup_{B_R(x)} |\underline{H}| \leq \Lambda, \quad \forall B_R(x) \subset U,$$

then set $h = 1$ we get

$$(1 + \epsilon)^{-n} J'(\rho) \leq \frac{d}{d\rho} (\rho^{-n} I(\rho)) + E_1(\rho) \leq J'(\rho),$$

where

$$E_1(\rho) = \rho^{-n} \int_U \rho^{-1} (x - \xi) \cdot \underline{H} \phi \left(\frac{r}{\rho} \right) d\mu = E(\rho) \rho^{-n} I(\rho)$$

and $E(\rho) \in \left[-(1 + \epsilon) \frac{\Lambda}{R}, (1 + \epsilon) \frac{\Lambda}{R} \right]$ for each $\rho \in (0, R)$.

Then multiplying $E^{F(\rho)}$, where $F(\rho) = \int_0^\rho E(t) dt \in \left[-(1 + \epsilon) \frac{\Lambda \rho}{R}, (1 + \epsilon) \frac{\Lambda \rho}{R} \right]$, we get

$$(1 + \epsilon)^{-n} e^{-(1+\epsilon)\Lambda} J'(\rho) \leq \frac{d}{d\rho} \left(e^{F(\rho)} \rho^{-n} I(\rho) \right) \leq e^{(1+\epsilon)\Lambda} J'(\rho).$$

Integrating over $[\sigma, \rho]$,

$$(1 + \epsilon)^{-n} e^{-(1+\epsilon)\Lambda} (J(\rho) - J(\sigma)) \leq \frac{e^{F(\rho)} I(\rho)}{\rho^n} - \frac{e^{F(\sigma)} I(\sigma)}{\sigma^n} \leq e^{(1+\epsilon)\Lambda} (J(\rho) - J(\sigma)).$$

Letting $\epsilon \rightarrow 0$,

$$\begin{aligned} \frac{e^{-\Lambda}}{\omega_n} \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} |D^\perp r|^2 d\mu &\leq \frac{e^{F(\rho)} \mu(B_\rho(\xi))}{\omega_n \rho^n} - \frac{e^{F(\sigma)} \mu(B_\sigma(\xi))}{\omega_n \sigma^n} \\ &\leq \frac{e^\Lambda}{\omega_n} \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} |D^\perp r|^2 d\mu. \end{aligned}$$

Thus we obtain the following theorem:

Theorem 2.7. *If U is open in $\mathbb{R}^{n+k}$, if $B_R(\xi) \subset U$ and V has generalized mean curvature vector $\underline{H}$ in U with $|\underline{H}| \leq \Lambda$, then*

$$e^{F(\rho)} \frac{\mu(B_\rho(\xi))}{\omega_n \rho^n} - e^{F(\sigma)} \frac{\mu(B_\sigma(\xi))}{\omega_n \sigma^n} = \frac{G(\sigma, \rho)}{\omega_n} \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} |D^\perp r|^2 d\mu$$

for all $0 < \sigma \leq \rho < R$, where $F(\rho) \in [-\frac{\Lambda\rho}{R}, \frac{\Lambda\rho}{R}]$ for all $0 < \rho < R$ and $G(\sigma, \rho) \in [e^{-\Lambda}, e^\Lambda]$ for all $0 < \sigma \leq \rho < R$.

By the way, in the formula

$$\rho \frac{d}{d\rho} \int_{B_\rho(\xi)} h |\nabla^M r|^2 d\mu = n \int_{B_\rho(\xi)} h d\mu + \int_{B_\rho(\xi)} (h \underline{H} + \nabla^M h) \cdot (x - \xi) d\mu,$$

we can directly compute the left hand side:

$$\frac{d}{d\rho} \int_{B_\rho(\xi)} h |\nabla^M r|^2 d\mu = \int_{\partial B_\rho(\xi)} h |\nabla^M r| d\nu$$

for $\mathcal{L}^1$ -a.e. $\rho \in (0, R)$ by the Co-area Formula, where $d\nu = \theta d\mathcal{H}^{n-1}$. Then

$$\rho \int_{\partial B_\rho(\xi)} h |\nabla^M r| d\mu = n \int_{B_\rho(\xi)} h d\mu + \int_{B_\rho(\xi)} (h \underline{H} + \nabla^M h) \cdot (x - \xi) d\mu.$$

Subtracting the equality above with $\sigma < \rho$ in place of ρ ,

$$\begin{aligned} & \int_{\partial B_\rho(\xi)} r h |\nabla^M r| d\nu - \int_{\partial B_\sigma(\xi)} r h |\nabla^M r| d\nu \\ &= n \int_{B_\rho(\xi) - B_\sigma(\xi)} h d\mu + \int_{B_\rho(\xi) - B_\sigma(\xi)} (h \underline{H} + \nabla^M h) \cdot (x - \xi) d\mu. \end{aligned}$$

Replacing h by $r^{-n}h$, and observing that

$$(x - \xi) \cdot \nabla^M r^{-n} = -nr^{-n} + nr^{-n} |D^\perp r|^2,$$

$$\begin{aligned} & \int_{\partial B_\rho(\xi)} r^{1-n} h |\nabla^M r| d\nu - \int_{\partial B_\sigma(\xi)} r^{1-n} h |\nabla^M r| d\nu \\ &= n \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} h d\mu + \int_{B_\rho(\xi) - B_\sigma(\xi)} (r^{-n} h \underline{H} + h \nabla^M r^{-n} + r^{-n} \nabla^M h) \cdot (x - \xi) d\mu \\ &= n \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} h d\mu + \int_{B_\rho(\xi) - B_\sigma(\xi)} (r^{-n} h \underline{H} + r^{-n} \nabla^M h) \cdot (x - \xi) d\mu \\ & \quad + \int_{B_\rho(\xi) - B_\sigma(\xi)} (-nhr^{-n} + nhr^{-n} |D^\perp r|^2) d\mu \\ &= n \int_{B_\rho(\xi) - B_\sigma(\xi)} hr^{-n} |D^\perp r|^2 d\mu + \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} (h \underline{H} + \nabla^M h) \cdot (x - \xi) d\mu. \end{aligned}$$

Now if we take $h : \mathbb{R}^{n+k} \rightarrow [0, 1]$ to be a homogeneous degree zero C^1 function of the variable $x - \xi$, and let $b \geq 1$ such that

$$|Dh(x - \xi)| \leq br^{-1}.$$

Because $Dh \cdot (x - \xi) = 0$,

$$(x - \xi) \cdot \nabla^M h = (x - \xi) \cdot p_{T_x M}(Dh) = p_{T_x M}(x - \xi) \cdot Dh = -rD^\perp r \cdot Dh,$$

then

$$|(x - \xi) \cdot \nabla^M h| \leq b |D^\perp r|.$$

We finally obtain

$$\begin{aligned} & \left| \rho^{1-n} \int_{\partial B_\rho(\xi)} h |\nabla^M r| d\nu - \sigma^{1-n} \int_{B_\sigma(\xi)} h |\nabla^M r| d\nu - n \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} |D^\perp r|^2 h d\mu \right| \\ & \leq \int_{B_\rho(\xi) - B_\sigma(\xi)} (r^{1-n} |\underline{H}| + br^{-n} |D^\perp r|) d\mu. \end{aligned}$$

Clearly it is a more precise estimate.

2.4.3 L^p -Generalized Mean Curvature

We continue to assume that $V = \underline{v}(M, \theta)$ is a rectifiable varifold in U with generalized mean curvature vector $\underline{H}$ in U , but now we assume that $\underline{H} \in L^p(p > n)$:

$$\left(R^{p-n} \int_{B_R(\xi)} |\underline{H}|^p d\mu_V \right)^{\frac{1}{p}} \leq \Lambda$$

for some $\Lambda > 0$. Then we have

$$\begin{aligned} & \left| \rho^{-n} \int_U \rho^{-1} (x - \xi) \cdot \underline{H} \phi\left(\frac{r}{\rho}\right) d\mu_V \right| \\ & \leq (1 + \epsilon) \rho^{-n} \|\underline{H}\|_{L^p(\mu_V \llcorner B_R(\xi))} (I(\rho))^{1 - \frac{1}{p}} \\ & \dots \end{aligned}$$

We have the following monotonicity formula:

$$\begin{aligned} & \left(e^{F(\rho)} \frac{\mu(B_\rho(\xi))}{\omega_n \rho^n} + \frac{1}{p} (e^{F(\rho)} - 1) \right) - \left(e^{F(\sigma)} \frac{\mu(B_\sigma(\xi))}{\omega_n \sigma^n} + \frac{1}{p} (e^{F(\sigma)} - 1) \right) \\ & = G(\sigma, \rho) \int_{B_\rho(\xi) - B_\sigma(\xi)} r^{-n} |D^\perp r|^2 d\mu_V, \end{aligned}$$

where

$$e^{-\kappa \Lambda} \leq G(\sigma, \rho) \leq e^{\kappa \Lambda}, \quad |F(\rho)| \leq \Lambda \kappa \left(\frac{\rho}{R} \right)^{1 - \frac{n}{p}}, \quad \kappa = \frac{p}{p - n}$$

for all $0 < \sigma < \rho < R$ provided $B_R(\xi) \subset U$.

3 Allard Regularity Theorem

3.1 Preliminaries

Definition 3.1. Suppose $V = v(M, \theta)$ is a rectifiable n -varifold with generalized mean curvature $\underline{H}$ in U . We call the tilt excess $E(\xi, \sigma, T)$ of V over a ball $B_\sigma(\xi) \subset U$ relative to a given n -dimensional subspace $T \subset \mathbb{R}^{n+k}$ defined by

$$E(\xi, \sigma, T) = \sigma^{-n} \int_{B_\sigma(\xi)} |p_{T_x M} - p_T|^2 d\mu_V(x).$$

(Here $|p_{T_x M} - p_T|^2 = \text{trace}(p_{T_x M} - p_T)^2$.)

Lemma 3.2. Suppose $B_\rho(y) \subset U$. Then for any n -dimensional subspace $T \subset \mathbb{R}^{n+k}$ and any $\gamma \in (0, 1)$, we have

$$\begin{aligned} & \rho^{-n} \int_{B_{\gamma\rho}(y)} |p_{T_x M} - p_T|^2 d\mu_V(x) \\ & \leq C\rho^{-n} \int_{B_\rho(y)} \left(\frac{\text{dist}(x, \xi + T)}{\rho} \right)^2 d\mu + C\rho^{2-n} \int_{B_\rho(y)} |\underline{H}|^2 d\mu_V, \end{aligned}$$

where C is a positive constant only depending on n, β .

Proof. We only need to prove the special case when $T = \mathbb{R}^n \times \{0\}$. By calculation,

$$E(y, \gamma\rho, T) = \frac{2}{(\gamma\rho)^n} \int_{B_{\gamma\rho}(y)} \sum_{j=1}^k |\nabla^M x^{n+j}|^2 d\mu_V.$$

Take $X = \zeta^2(x)x'$, $\zeta \in C_c^1(U)$, where $x' = (0, \dots, 0, x^{n+1}, \dots, x^{n+k})$. We have

$$\text{div}_M x' = \sum_{j=1}^{n+k} e_j \cdot \nabla^M x' = \sum_{j=n+1}^{n+k} e_j \cdot p_{T_x M}(e_j) = \sum_{j=n+1}^{n+k} e^{jj},$$

where $\{e^{ij}\}$ denotes the matrix of $p_{T_x M}$. Now using the definition of generalized mean curvature vector,

$$\int_U (2\zeta \nabla^M \zeta \cdot x' + \zeta^2 \text{div}_M x') d\mu_V = - \int_U \zeta^2 x' \cdot \underline{H} d\mu_V,$$

or

$$\int_U \sigma \zeta^2 d\mu_V = - \int_U \left(2\zeta \sum_{i=n+1}^{n+k} \sum_{j=1}^{n+k} x^i e^{ij} \partial_j \zeta - \zeta^2 x' \cdot \underline{H} \right) d\mu_V,$$

with

$$\sigma = \sum_{j=n+1}^{n+k} e^{jj} = \frac{1}{2} \sum_{i,j=1}^{n+k} (e^{ij} - e^{ji})^2 = \frac{1}{2} |p_{T_x M} - p_{\mathbb{R}^n \times \{0\}}|^2,$$

where $\{\epsilon^{ij}\}$ denotes the matrix of $p_{\mathbb{R}^n \times \{0\}}$. Because if $i > n$ then $\epsilon^{ij} = 0$, so

$$\begin{aligned} \int_U \sigma \zeta^2 d\mu_V &= - \int_U \left(2\zeta \sum_{i=n+1}^{n+k} \sum_{j=1}^{n+k} x^i (e^{ij} - \epsilon^{ij}) \partial_j \zeta + \zeta^2 x' \cdot \underline{H} \right) d\mu_V \\ &\leq \int_U \left(2\sqrt{2}\sigma |x'| |\nabla \zeta| \zeta + |x'| |\underline{H}| \zeta^2 \right) d\mu_V \\ &\leq \int_U \left(\frac{1}{2} \sigma \zeta^2 + 4|x'|^2 |\nabla \zeta|^2 + |x'| |\underline{H}| \zeta^2 \right) d\mu_V, \end{aligned}$$

so

$$\int_U \sigma \zeta^2 d\mu_V \leq 8 \int_U \left(|x'|^2 |\nabla \zeta|^2 + |x'| |\underline{H}| \zeta^2 \right) d\mu_V.$$

We only need to choose ζ such that $\zeta = 1$ in $B_{\gamma\rho}(0)$, $\zeta = 0$ outside $B_\rho(0)$ and $|\nabla \zeta| \leq \frac{2}{(1-\gamma)\rho}$, then

$$\begin{aligned} \int_U \sigma \zeta^2 d\mu_V &\leq \frac{32}{(1-\gamma)^2 \rho^2} \int_U |x'|^2 d\mu_V + 8 \int_U |x'| |\underline{H}| d\mu_V \\ &\leq \left(\frac{32}{(1-\gamma)^2} + 4 \right) \cdot \frac{1}{\rho^2} \int_U |x'|^2 d\mu_V + 4\rho^2 \int_U |\underline{H}|^2 d\mu_V. \end{aligned}$$

Because

$$\int_{B_{\gamma\rho}(y)} |p_{T_x M} - p_T|^2 d\mu_V(x) = 4 \int_{B_{\gamma\rho}(y)} \sigma d\mu_V,$$

we obtain what we want to prove. $\square$

Next, we shall assume that there exists a constant $\delta \in (0, \frac{1}{4}]$, whose value will be specific later and $\mu = \mu_V = \mathcal{H}^n \llcorner \theta$ satisfying the following (*) condition:

$$\begin{cases} \mu \geq \theta \text{ } \mu\text{-a.e.}, \\ 0 \in \text{spt} V, \ B_\rho(0) \subset U, \\ \omega_n^{-1} \rho^{-n} \mu(B_\rho(0)) \leq 1 + \delta, \\ \left(\rho^{p-n} \int_{B_\rho(0)} |\underline{H}|^p d\mu \right)^{\frac{1}{p}} \leq \delta. \end{cases}$$

Without loss of generality, we assume $\theta(y) = \Theta^n(\mu_V, y)$ for all $y \in B_1(0)$, and $\Theta(\mu, x)$ is an upper semi-continuous function of x .

By the monotonicity formula, if $y \in \text{spt} V \cap B_{2\delta\rho}(0)$, $\sigma \in (0, (1-2\delta)\rho]$, then

$$1 - C\delta \leq \frac{1}{\omega_n \sigma^n} \mu(B_\sigma(y)) \leq (1 + C\delta) \frac{1}{\omega_n (1-2\delta)^n \rho^n} \mu(B_{(1-2\delta)\rho}(y)) \leq 1 + C\delta.$$

Thus we call take a sufficiently small $\delta_0 > 0$ only depending on n, k, p , such that for any $\delta \leq \delta_0$,

$$\frac{1}{2} \leq 1 - C\delta \leq \frac{\mu(B_\sigma(y))}{\omega_n \sigma^n} \leq 1 + C\delta \leq 2$$

for any $\sigma \in (0, (1 - 2\delta)\rho]$, $y \in \text{spt}V \cap B_{2\sigma\rho}(0)$. Also, we have

$$\int_{B_{(1-2\delta)\rho}(y)} |x - y|^{-n} \left| \left(\frac{x - y}{|x - y|} \right)^\perp \right|^2 d\mu \leq C\delta$$

for any $y \in \text{spt}\mu \cap B_{2\delta\rho}(0)$.

The lemma below shows that points near every fixed point $y \in \text{spt}\mu_V$ remains close to a subspace.

Lemma 3.3 (Affine Approximation Lemma). *If $\delta \in (0, \frac{1}{16}]$ and $(*)$ holds, then for each $y \in \text{spt}\mu_V \cap B_{2\delta\rho}(0)$ and each $\sigma \in (0, 4\delta\rho]$,*

$$\sup_{x \in \text{spt}\mu_V \cap B_\sigma(y)} \text{dist}(x, y + T) \leq C\delta^{\frac{1}{2n+2}}\sigma,$$

where C is a positive constant only depending on n, k, p , and T is a n -dimensional subspace of $\mathbb{R}^{n+k}$ only depending on y, σ .

Proof. For any fixed $\sigma \in (0, 4\delta\rho]$ and $y \in \text{spt}V \cap B_{2\delta\rho}(0)$, and without loss of generality, suppose $\sigma = 1, y = 0$. Because $B_{2\sigma}(0) \subset B_{8\delta\rho}(0) \subset B_{(1-4\delta)\rho}(0)$, for any $r \in (0, 2]$ and $y \in \text{spt}\mu \cap B_1(0)$,

$$\frac{1}{2} \leq \frac{\mu(B_r(y))}{\omega_n \sigma^n} \leq 2,$$

and

$$\int_{B_2(y)} |p_{(T_x M)^\perp}(x - y)|^2 d\mu \leq 2^{n+2} \int_{B_2(y)} |p_{(T_x M)^\perp}(x - y)|^2 |x - y|^{-(n+2)} d\mu \leq C\delta.$$

Take any $\alpha \in (0, 1)$ whose value will be exact later. We can find finite $\{y_j\}_{j=1}^N \subset \text{spt}\mu \cap B_1(0)$ such that

$$\text{spt}\mu \cap B_1(0) \subset \bigcup_{j=1}^n B_{\delta^\alpha}(y_j),$$

and $\{B_{\delta^\alpha}(y_j)\}_{j=1}^N$ are pairwise disjoint. First, the number N can be bounded: since

$$\frac{\omega_n \delta^{\alpha n}}{2^{n+1}} \leq \mu(B_{\delta^\alpha}(y_j)), \quad j = 1, \dots, N,$$

we obtain

$$\frac{\omega_n \delta^{\alpha n} N}{2^{n+1}} \leq \sum_{j=1}^N \mu(B_{\delta^\alpha}(y_j)) = \mu\left(\bigcup_{j=1}^M B_{\delta^\alpha}(y_j)\right) \leq \mu(B_2(0)) \leq 2^{n+1} \omega_n,$$

which gives $N \leq 4^{n+1} \delta^{-\alpha n}$. Noting that $B_1(0) \subset B_2(y_j)$ for each j , we have

$$\int_{B_1(0)} \sum_{j=1}^N |p_{(T_x M)^\perp}(x - y_j)|^2 d\mu \leq CN\delta \leq C\delta^{1-\alpha n}.$$

Therefore, for any given $k \geq 1$, there exists a μ -measurable subset $F(k) \subset \text{spt}\mu \cap B_1(0)$ satisfying $\mu((\text{spt}\mu \cap B_1(0)) - F(k)) \leq \frac{1}{k}$, and

$$\sum_{j=1}^N |p_{(T_x M)^\perp}(x - y_j)|^2 \leq Ck\delta^{1-\alpha n} \text{ in } F(k).$$

Since the measure of the sets of "bad points" is less than $\frac{1}{k}$, considering $\mu(B_{\delta^\alpha}(0)) \geq C\delta^{-\alpha n}$, in order to guarantee there exists at least one "good point" in $B_{\delta^\alpha}(0)$, we simply need to take $k = 2C\delta^{-\alpha n}$. Therefore, there exists $x_0 \in \text{spt}\mu \cap B_{\delta^\alpha}(0)$ such that

$$\sum_{j=1}^N |p_{(T_{x_0} M)^\perp}(x_0 - y_j)|^2 \leq C\delta^{1-2\alpha n},$$

hence

$$|p_{(T_{x_0} M)^\perp}(x_0 - y_j)| \leq C\delta^{\frac{1}{2}-\alpha n}, \quad \forall j = 1, \dots, N.$$

Since $|x_0| \leq \delta^\alpha$,

$$|p_{(T_{x_0} M)^\perp} y_j| \leq C \left(\delta^{\frac{1}{2}-\alpha n} + \delta^\alpha \right), \quad \forall j = 1, \dots, N.$$

To obtain optimal results, we take $\frac{1}{2} - \alpha n = \alpha \iff \alpha = \frac{1}{2n+2}$. Thus all of $\{y_j\}_{j=1}^N$ are in the $C\delta^{\frac{1}{2n+2}}$ -neighborhood of the subspace $T_0 = T_{x_0} M$. And since for any $y \in B_1(0)$, there exists a y_j such that $|y_j - y| \leq \delta^{\frac{1}{2n+2}}$, we finally obtain the estimate for all points in $B_1(0)$. $\square$

3.2 Lipschitz Approximation

Lemma 3.4 (Lipschitz Approximation Theorem). *Let $L \in (0, 1]$ be given, and there is a constant $\beta = \beta(n, k, p) \in (0, \frac{1}{16}]$ such that if $0 < \delta \leq (\beta L)^{2n+2}$, $\rho = 1$, (*) holds, and if $T(0, 4\delta) = \mathbb{R}^n \times \{0\}$, then there exists a Lipschitz function $f : B_\delta^n(0) \rightarrow \mathbb{R}^k$ with*

$$\text{Lip} f \leq 1, \sup |f| \leq C\delta^{\frac{1}{2n+2}}$$

and

$$\begin{aligned} & \delta^{-n} (\mu_V(B_\delta(0) \cap (\text{spt}\mu_V - F)) + \mathcal{H}^n(B_\delta(0) \cap (F - \text{spt}\mu_V))) \\ & \leq \frac{C}{L^2 \delta^n} \int_{B_{3\delta}(0)} |p_{T_x M} - p_{\mathbb{R}^n \times \{0\}}|^2 d\mu_V \leq \frac{C}{L^2} \delta^{\frac{1}{n+1}}, \end{aligned}$$

where $F = \text{graph} f$ and $C > 0$ is a constant only depending on n, k, p .

Proof. Let $\beta \in (0, \frac{1}{2}]$ whose value will be specific later. Let

$$G = \{y \in \text{spt}\mu \cap B_{2\delta}(0) : \sup_{\sigma \in (0, \delta]} \sigma^{-n} \int_{B_\sigma(y)} |p_{T_x M} - p_{T_0}|^2 d\mu \leq \beta^2 L^2\}.$$

Then for any $y \in (\text{spt}\mu \cap B_{2\delta}(0)) - G$, there exists $\sigma = \sigma_y \in (0, \delta]$ such that

$$\sigma^{-n} \int_{B_\sigma(y)} |p_{T_x M} - p_T|^2 d\mu > \beta^2 L^2,$$

and we have $(\text{spt}\mu \cap B_{2\delta}(0)) - G \subset \bigcup_{y \in (\text{spt}\mu \cap B_{2\delta}(0)) - G} B_{\sigma_y}(y)$. By the $5r$ -covering lemma, we can pick pairwise disjoint ball $B_{\delta_j}(y_j)$ such that $y_j \in (\text{spt}\mu \cap B_{2\delta}(0)) - G$, $\sigma_j \in (0, \delta_j]$ satisfying

$$\sigma_j^{-n} \int_{B_{\sigma_j}(y_j)} |p_{T_x M} - p_T|^2 d\mu > \beta^2 L^2,$$

and

$$(\text{spt}\mu \cap B_{2\delta}(0)) - G \subset \bigcup_j B_{5\sigma_j}(y_j).$$

Then

$$\begin{aligned} \beta^2 L^2 \mu(B_{2\delta}(0) - G) &\leq \beta^2 L^2 \sum_j \mu(B_{5\delta_j}(y_j)) \leq C \beta^2 L^2 \sum_j \sigma_j^n \\ &\leq \int_{\bigcup_j B_{\sigma_j}(y_j)} |p_{T_x M} - p_{T_0}|^2 d\mu \leq C \int_{B_{3\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu, \end{aligned}$$

so

$$\mu(B_{2\delta}(0) - G) \leq \frac{C}{\beta^2 L^2} \int_{B_{3\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu.$$

Next we shall prove that G is contained in the graph of a Lipschitz function. For any distinct points $y_1, y_2 \in G$, let $\sigma = |y_1 - y_2|$, so $\sigma \leq 4\delta$. Observe that

$$\sigma^{-n} \int_{B_{\frac{\sigma}{4}}(y_1)} |p_{T_x M} - p_{T_0}|^2 \leq 4^n \beta^2 L^2,$$

and

$$\sigma^{-n} \int_{B_{\frac{\sigma}{4}}(y_1)} |p_{T_x M} - p_{T(y_1, \sigma)}|^2 \leq C \delta^{\frac{1}{n+1}} \leq C \beta^2 L^2.$$

Since σ is arbitrary, we get

$$|p_{T_0} - p_{T(y_1, \sigma)}|^2 \leq C \beta L,$$

so

$$\begin{aligned} |p_{T_0}^\perp(y_1 - y_2)| &= |(p_{T(y, \sigma)}^\perp + (p_{T_0} - p_{T(y_1, \sigma)}^\perp)(y_1 - y_2)| \\ &\leq \text{dist}(y_2, y_1 + T(y_1, \sigma)) + |p_{T_0} - p_{T(y_1, \sigma)}| \sigma \leq C \beta L \sigma. \end{aligned}$$

Thus,

$$|Q(y_2 - y_1)| \leq C \beta L |y_1 - y_2| \leq C \beta L (|Q(y_1 - y_2)| + |p(y_1 - y_2)|),$$

where P, Q are the projections of $y = (y^1, \dots, y^n)$ onto its first n and last k coordinates respectively. Assuming $C\beta \leq \frac{1}{2}$, then

$$|Q(y_1) - Q(y_2)| \leq C\beta L |p(y_1) - P(y_2)|.$$

Consider the arbitrariness of $y_1, y_2 \in G$, we obtain that G is contained in the graph of a Lipschitz function $f : G \cap B_\delta(0) \rightarrow \mathbb{R}^k$ with Lipschitz constant not greater than $C\beta L$, provided we choose $\beta = \beta(n, k, p)$ such that $C\beta \leq \frac{1}{2}$. Also, $\sup |f| \leq C\delta^{\frac{1}{2n+2}}\sigma$, so by the Lipschitz Extension Theorem, f extends to a Lipschitz function $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}^k$ with

$$\begin{cases} G \cap B_{2\delta}(0) \subset \text{graph} f, \\ \text{Lip} f \leq C\beta L, \\ \sup_{\mathbb{R}^n} |f| \leq C\beta \delta^{\frac{1}{2n+2}}\sigma. \end{cases}$$

Thus we get $\text{Lip} f \leq L$ if we demand $C\beta \leq 1$. Also, for $F = \text{graph} f$, we have

$$\delta^{-n} \mu(B_{2\delta}(0) - F) \leq \frac{C}{\beta^2 L^2 \delta^n} \int_{B_{3\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu \leq \frac{C\delta^{\frac{1}{n+1}}}{L^2}.$$

Therefore, we only need to prove that

$$\mathcal{H}^n(B_\delta(0) \cap (F - \text{spt} \mu)) \leq \frac{C}{L^2} \int_{B_{3\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu.$$

Take any $\eta \in B_\delta(0) \cap (F - \text{spt} \mu)$ and let

$$\sigma = \frac{3}{2} \text{dist}(\eta, \text{spt} \mu) \leq \frac{3}{2} \delta.$$

By monotonicity identity,

$$\begin{aligned} \sigma^{-n} \mu(B_\sigma(\eta)) &= \sigma^{-n} \mu(B_\sigma(\eta)) - \left(\frac{\sigma}{2}\right)^{-n} \mu(B_{\frac{\sigma}{2}}(\eta)) \\ &\leq C \int_{B_\sigma(\eta) - B_{\frac{\sigma}{2}}(\eta)} |x - \eta|^{-n} \left| p_{(T_x M)^\perp} \left(\frac{x - \eta}{|x - \eta|} \right) \right|^2 d\mu + C\delta \\ &\leq C\sigma^{-(n+2)} \int_{B_\sigma(\eta)} |p_{(T_x M)^\perp}(x - \eta)|^2 d\mu + C\delta. \end{aligned}$$

Because $\text{spt} \mu \cap B_{\frac{2\sigma}{3}}(\eta) \neq \emptyset$, we have

$$\mu(B_\sigma(\eta)) \geq \mu(B_{\frac{\sigma}{3}}(y')) \geq \frac{\omega_n \sigma^n}{3^{n+1}}$$

where $y' \in \text{spt} \mu$ such that $\text{dist}(\eta, y') = \frac{\sigma}{3}$. Also by $C\delta \leq C(\beta L)^{2n+2} \leq C\beta^{2n+2}$, hence for small $\beta = \beta(n, k, p)$, we obtain

$$\begin{aligned} \sigma^n &\leq C \int_{B_\sigma(\eta)} \left| p_{(T_x M)^\perp} \left(\frac{x - \eta}{\sigma} \right) \right|^2 d\mu \\ &\leq C \left(\int_{B_\sigma(\eta)} \left| p_{T_0^\perp} \left(\frac{x - \eta}{\sigma} \right) \right|^2 d\mu + \int_{B_\sigma(\eta)} |p_{T_x M} - p_{T_0}|^2 d\mu \right). \end{aligned}$$

We assert that $\sigma \in (0, \delta)$, which show that points in $F - \text{spt}\mu$ cannot be too far from $\text{spt}\mu$. For if $\sigma \in [\delta, \frac{3}{2}\delta]$, we get

$$\int_{B_{3\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu \leq C\delta^n \delta^{\frac{1}{n+1}} \leq C\sigma^n \delta^{\frac{1}{n+1}},$$

$$\int_{B_{4\delta}(0)} \left| p_{T_0^\perp} \left(\frac{x - \eta}{\sigma} \right) \right|^2 d\mu \leq \int_{B_{4\delta}(0)} \frac{C}{\sigma^2} \cdot \delta^{\frac{1}{n+1}} \sigma^2 d\mu \leq C\delta^n \cdot \delta^{\frac{1}{n+1}} \leq C\sigma^n \delta^{\frac{1}{n+1}},$$

then since $B_\sigma(\eta) \subset B_{\frac{3}{2}\delta}(\eta) \subset B_{3\delta}(0)$,

$$\sigma^n \leq C\sigma^n \delta^{\frac{1}{n+1}} \leq C\sigma^n \beta^2 L^2.$$

Therefore, if we take β sufficiently small, then we get a contradiction. So $\sigma \in (0, \delta)$. We also have

$$\begin{aligned} & \int_{B_\sigma(\eta)} \left| p_{T_0^\perp} \left(\frac{x - \eta}{\sigma} \right) \right|^2 d\mu + \int_{B_\sigma(\eta)} |p_{T_x M} - p_{T_0}|^2 d\mu \\ & \leq \int_{B_\sigma(\eta) \cap F} \left| p_{T_0^\perp} \left(\frac{x - \eta}{\sigma} \right) \right|^2 d\mu + C\mu(B_\sigma(\eta) - F) + \int_{B_\sigma(\eta)} |p_{T_x M} - p_{T_0}|^2 d\mu. \end{aligned}$$

Since for $x, y \in F \cap B_\sigma(\eta)$ we have $\left| p_{T_0|_{\text{perp}}} \left(\frac{x-y}{\sigma} \right) \right| \leq C\beta$ and $\mu(B_\sigma(\eta)) \leq 2\omega_n \sigma^n$, we have

$$\sigma^n \leq C \left(\beta\sigma + \mu(B_\sigma(\eta) - F) + \int_{B_\sigma(\eta)} |p_{T_x M} - p_{T_0}|^2 d\mu \right).$$

If we choose β appropriately, we can promise that $C\beta \leq \frac{1}{2}$, and hence we obtain

$$\sigma^n \leq C \left(\mu(B_\sigma(\eta) - F) + \int_{B_\sigma(\eta)} |p_{T_x M} - p_{T_0}|^2 d\mu \right)$$

for some $\sigma \in (0, \delta]$.

Now because the collection of such balls $B_\sigma(\eta)$ cover $B_\delta(0) \cap (F - \text{spt}\mu)$, by the 5r-covering lemma, we can find a pairwise disjoint collection $B_{\sigma_j}(\eta_j)$ of such balls with

$$\sigma_j^n \leq C \left(\mu(B_{\sigma_j}(\eta_j) - F) + \int_{B_{\sigma_j}(\eta_j)} |p_{T_x M} - p_{T_0}|^2 d\mu \right)$$

for each j and $B_\delta(0) \cap (F - \text{spt}\mu) \subset \bigcup_j B_{5\sigma_j}(\eta_j)$, where F is the graph of the Lipschitz function f with $\text{Lip} f \leq 1$, so we have $\mathcal{H}^n(B_{5\sigma_j}(\eta_j) \cap F) \leq C\sigma_j^n$ for

each j , where $C = C(n)$. Hence,

$$\begin{aligned}
\mathcal{H}^n(B_\delta(0) \cap (F - \text{spt}\mu)) &\leq \mathcal{H}^n\left(F \cap \left(\bigcup_j B_{5\sigma_j}(\eta_j)\right)\right) \leq \sum_j \mathcal{H}^n(F \cap B_{5\sigma_j}(\eta_j)) \\
&\leq C \sum_j \sigma_j^n \leq C \sum_j \left(\mu(B_{\sigma_j}(\eta_j) - F) + \int_{B_{\sigma_j}(\eta_j)} |p_{T_x M} - p_{T_0}|^2 d\mu\right) \\
&\leq C \left(\mu\left(\bigcup_j B_{\sigma_j}(\eta_j) - F\right) + \int_{\bigcup_j B_{\sigma_j}(\eta_j)} |p_{T_x M} - p_{T_0}|^2 d\mu\right) \\
&\leq C \left(\mu(B_{2\delta}(0) - F) + \int_{B_{2\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu\right) \\
&\leq CL^{-2} \int_{B_{3\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu + C \int_{B_{2\delta}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu.
\end{aligned}$$

□

Corollary 3.5. *There exists $\beta = \beta(n, k, p) \in (0, \frac{1}{4}]$ such that if the lemma above holds, and if*

$$\sup_{\sigma \in (0, \sigma]} \sigma^{-n} \int_{B_\sigma(y)} |p_{T_x M} - p_{T_0}|^2 d\mu \leq \beta^2 L^2$$

for every $y \in \text{spt}\mu \cap B_{2\delta}(0)$, then

$$\text{spt}\mu \cap B_\delta(0) = \text{graph} f \cap B_\delta(0)$$

for some Lipschitz map $f : \mathbb{R}^n \rightarrow \mathbb{R}^k$ with $\text{Lip} f \leq L$, $\sup |f| \leq C\delta^{\frac{1}{2n+2}}$.

Proof. We only need to show that in the prove above, G is in fact $\text{spt}\mu_V \cap B_{2\delta}(0)$. For any $\eta \in B_\delta(0) \cap (\text{graph} f - \text{spt}\mu_V)$, we have

$$\sigma^n \leq C \int_{B_\sigma(\eta)} |p_{T_x M} - p_{T_0}|^2 d\mu \leq C\beta^n \sigma^n$$

for some $\sigma \in (0, \delta]$, which is impossible when β is sufficiently small. □

3.3 Harmonic Approximation

Lemma 3.6. *Given any $\epsilon > 0$, there exists a constant $\delta = \delta(n, \epsilon) > 0$ such that if $f \in W^{1,2}(B)$, $B = B_1(0)$ satisfies*

$$\int_B |\nabla f|^2 \leq 1, \quad \left| \int_B \nabla f \cdot \nabla \zeta d\mathcal{L}^n \right| \leq \delta \sup |\nabla \zeta|$$

for every $\zeta \in C_c^\infty(B)$, then there exists a harmonic function u on B such that

$$\int_B |\nabla u|^2 \leq 1,$$

$$\int_B (u - f)^2 \leq \epsilon.$$

Proof. Suppose the lemma is false, then there exist $\epsilon > 0$ and $\{f_k\} \subset W^{1,2}(B)$ such that

$$\left| \int_B \nabla f_k \cdot \nabla \zeta d\mathcal{L}^n \right| \leq \frac{1}{k} \sup |\nabla \zeta|, \quad \forall \zeta \in C_c^\infty(B),$$

$$\int_B |\nabla f_k|^2 \leq 1,$$

but

$$\int_B (f_k - u)^2 > \epsilon$$

for every u harmonic on B with $\int_B |\nabla u|^2 \leq 1$. Let $\lambda_k = \omega_n^{-1} \int_B f_k d\mathcal{L}^n$, then by Poincare inequality,

$$\int_B |f_k - \lambda_k|^2 \leq C \int_B |\nabla f_k|^2 \leq C,$$

and by Rellich's theorem, there exists a subsequence which we still name it $\{f_k\}$ such that $f_k - \lambda_k \rightarrow w$ in $L^2(B)$ strongly and $\nabla f_k \rightarrow \nabla w$ weakly in $L^2(B)$, where $w \in W^{1,2}(B)$ and $\int_B |\nabla w|^2 \leq 1$. Obviously we have

$$\int_B \nabla w \cdot \nabla \zeta d\mathcal{L}^n = \lim_{k \rightarrow \infty} \int_B \nabla f_k \cdot \nabla \zeta d\mathcal{L}^n = 0, \quad \forall \zeta \in C_c^\infty(B).$$

So we get that w is harmonic in B and

$$\int_B |f_k - w - \lambda_k|^2 \rightarrow 0.$$

Since $w + \lambda_k$ is harmonic, this leads to a contradiction. $\square$

3.4 Tilt-Excess Decay Lemma

We use the notation:

$$E(\xi, \sigma, T) = \sigma^{-n} \int_{B_\sigma(\xi)} |p_{T_x M} - p_T|^2 d\mu_V(x),$$

$$E_*(\xi, \sigma, T) = \max \left\{ E(\xi, \sigma, T), \frac{1}{\delta} \left(\sigma^{p-n} \int_{B_\sigma(\xi)} |\underline{H}|^p d\mu \right)^{\frac{2}{p}} \right\},$$

where δ is a sufficiently small positive constant in condition (*).

Theorem 3.7 (Tilt-Excess Decay Theorem). *There exist constants $\eta = \eta(n, k, p)$, $\delta_0 = \delta_0(n, k, p) \in (0, \frac{1}{16}]$ such that if $0 < \delta \leq \delta_0$ and (*) holds, if $\sigma \in (0, \delta\rho]$, $\xi \in \text{spt}\mu_V \cap B_{\delta\rho}(0)$, and if T is any n -dimensional subspace of $\mathbb{R}^{n+k}$, then*

$$E_*(\xi, \eta\sigma, S) \leq \eta^{2(1-\frac{n}{p})} E_*(\xi, \sigma, T)$$

for some n -dimensional subspace $S \subset \mathbb{R}^{n+k}$.

Remark 3.8. For any such $S \subset \mathbb{R}^{n+k}$, automatically, we have

$$\frac{1}{(\eta\sigma)^n} \int_{B_{\eta\sigma}(\xi)} |p_{T_x M} - p_T|^2 d\mu \leq \frac{1}{\eta^n} E(\xi, \sigma, T),$$

and by the Tilt-Excess Decay Theorem,

$$\frac{1}{(\eta\sigma)^n} \int_{B_{\eta\sigma}(\xi)} |p_{T_x M} - p_S|^2 d\mu \leq E_*(\xi, \sigma, T),$$

then we get

$$\frac{1}{(\eta\sigma)^n} \int_{B_{\eta\sigma}(\xi)} |p_T - p_S|^2 d\mu \leq \frac{C}{\eta^n} E_*(\xi, \sigma, T).$$

Let $\sigma \rightarrow 0$, we finally obtain

$$|p_S - p_T|^2 \leq \frac{C}{\eta^n} E_*(\xi, \sigma, T).$$

Proof. Without loss of generality, set $\xi = 0$, $T = \mathbb{R}^n \times \{0\}$. Let $T_0 = T(0, 2\sigma)$, then we have

$$E(0, \sigma, T_0) \leq C\delta^{\frac{1}{n+1}}.$$

If $E(0, \sigma, T) > C\delta^{\frac{1}{n+1}}$, then taking $S = T_0$, we get what we want. So we can assume that $E(0, \sigma, T) < C\delta^{\frac{1}{n+1}}$. Then we get

$$|p_T - p_{T_0}| \leq C\delta^{\frac{1}{2n+2}}.$$

Hence, since $\sup_{x \in \text{spt}\mu \cap B_\sigma(0)} \text{dist}(x, T_0) \leq C\delta^{\frac{1}{2n+2}}\sigma$, we also have $\sup_{x \in \text{spt}\mu \cap B_\sigma(0)} \text{dist}(x, T_0)$.

Since $T = \mathbb{R}^n \times \{0\}$, this is equivalent to

$$\sup_{B_\sigma(0) \cap \text{spt}\mu} \sum_{j=1}^k |x^{n+j}| \leq C\delta^{\frac{1}{2n+2}}\sigma.$$

By the Lipschitz Approximation Lemma, there exists a Lipschitz function $f : B_{\frac{\sigma}{3}}^n(0) \rightarrow \mathbb{R}^k$ with

$$\begin{cases} \text{Lip} f \leq 1, \\ \sup |f| \leq C\delta^{\frac{1}{2n+2}}\sigma, \\ \mu(\text{spt}\mu \cap B_{\frac{\sigma}{3}}(0) - F) + \mathcal{H}^n(F \cap B_{\frac{\sigma}{3}}(0) - \text{spt}\mu) \leq CE_0\sigma^n, \end{cases}$$

where $F = \text{graph} f$, $E_0 = E_*(0, \sigma, T)$. Take δ sufficiently small such that $C\delta^{\frac{1}{2n+2}} \leq \frac{1}{5}$, then

$$\text{spt}\mu \cap B_\sigma(0) \cap \left(B_{\frac{\sigma}{4}}^n(0) \times \mathbb{R}^k \right) \subset B_{\frac{\sigma}{4}}^n(0) \times B_{\frac{\sigma}{5}}^k(0) \subset B_{\frac{\sigma}{5}}(0).$$

We are now going to prove that each component of the Lipschitz function f can be approximated by a harmonic function. Noting that by the definition of generalized mean curvature,

$$\int_M \operatorname{div}_M X d\mu_V = - \int_M X \cdot \underline{H} d\mu_V, \quad \forall X \in C_c^1 \left(B_{\frac{\delta}{3}}(0), \mathbb{R}^{n+k} \right),$$

we simply take $X = \zeta e_{n+j}$, $\zeta \in C_c^1 \left(B_{\frac{\delta}{3}}(0) \right)$ for any fixed $j = 1, \dots, k$, then

$$\int_M (\nabla^M x^{n+j}) \cdot \nabla^M \zeta d\mu = - \int_M e_{n+j} \cdot \underline{H} \zeta d\mu.$$

Since $x^{n+j} = \tilde{f}^j(x)$ on $M \cap F$ (where $\tilde{f}$ is defined by $\tilde{f}^j(x^1, \dots, x^{n+k}) = f^j(x^1, \dots, x^n)$), we have

$$\nabla^M x^{n+j} = \nabla^M \tilde{f}^j(x)$$

for μ -a.e. $x \in M \cap F$. Then

$$\int_{M \cap F} \nabla^M \tilde{f}^j \cdot \nabla^M \zeta d\mu = - \int_{M-F} \nabla^M x^{n+j} \cdot \nabla^M \zeta d\mu - \int_M e_{n+j} \cdot \underline{H} \zeta d\mu.$$

Since we have

$$\int_{B_\sigma(\xi)} |\underline{H}| d\mu \leq \left(\int_{B_\sigma(\xi)} |\underline{H}|^p d\mu \right)^{\frac{1}{p}} (\mu(B_\sigma(\xi)))^{1-\frac{1}{p}} \leq C \delta^{\frac{1}{2}} E_0^{\frac{1}{2}} \sigma^{n-1},$$

we obtain

$$\begin{aligned} \left| \sigma^{-n} \int_{M \cap F} \nabla^M \tilde{f}^j \cdot \nabla^M \zeta d\mu \right| &\leq C E_0 \sup |\nabla \zeta| + C \delta^{\frac{1}{2}} E_0^{\frac{1}{2}} \sigma^{-1} \sup |\zeta| \\ &\leq C \left(E_0 + \delta^{\frac{1}{2}} E_0^{\frac{1}{2}} \right) \sup |\nabla \zeta|. \end{aligned}$$

And trivially by definition we have

$$\sigma^{-n} \int_{M \cap F \cap B_{\frac{\delta}{3}}(0)} \left| \nabla^M \tilde{f} \right|^2 d\mu \leq 2E_0.$$

To apply the Harmonic Approximation Theorem here, we also need information of $\nabla^M f \cdot \nabla \zeta$ and $|\nabla f|^2$. For any $\zeta \in C_c^1 \left(B_{\frac{\delta}{3}}(0) \right)$, let $\tilde{\zeta}(x^1, \dots, x^{n+k}) = \zeta(x^1, \dots, x^n)$, so $\operatorname{spt} \tilde{\zeta} \subset B_{\frac{\delta}{3}}(0) \times \mathbb{R}^k$. Since $\operatorname{spt} \mu \cap B_\sigma(0) \cap \left(B_{\frac{\delta}{4}}(0) \times \mathbb{R}^k \right) \subset B_{\frac{\delta}{3}}(0)$, we can find a function $\hat{\zeta} \in C_0^1 \left(B_{\frac{\delta}{3}}(0) \right)$ which agrees with $\tilde{\zeta}$ in a neighborhood of $\operatorname{spt} \mu \cap \operatorname{spt} \tilde{\zeta}$. So we can use $\hat{\zeta}$ in place of ζ in the above discussion. Also,

$$\nabla^M \tilde{f}^j \cdot \nabla^M \tilde{\zeta} = \sum_{i,l=1}^n e^{il} \partial_i \tilde{f} \partial_l \tilde{\zeta} = \nabla \tilde{f} \cdot \nabla \tilde{\zeta} - \sum_{i,l=1}^n (\delta_{il} - e^{il}) \partial_i \tilde{f}^j \partial_l \tilde{\zeta},$$

where $\{e^{il}\}_{i,l=1}^{n+k}$ is the matrix of $p_{T_x M}$, and the maximum eigenvalue of $\{\delta_{il} - e^{il}\}_{i,l=1}^n$ is not greater than $\text{trace}\{\delta_{il} - e^{il}\}_{i,l=1}^n$. Then

$$\begin{aligned} \left| \sum_{i,l=1}^n (\delta_{il} - e^{il}) \partial_i \tilde{f} \partial_l \tilde{\zeta} \right| &\leq \left(n - \sum_{i=1}^n e^{ii} \right) |\partial_i \tilde{f}| |\partial_l \tilde{\zeta}| \\ &= \left(n - \sum_{i=n+1}^{n+k} e^{ii} \right) |\partial_i \tilde{f}| |\partial_l \tilde{\zeta}| \\ &= \frac{1}{2} |p_{T_x M} - p_T|^2 |\partial_i \tilde{f}| |\partial_l \tilde{\zeta}|. \end{aligned}$$

So since $|\nabla \tilde{f}^j| \leq 1$, we obtain

$$\left| \nabla^M \tilde{f} \cdot \nabla^M \tilde{\zeta} - \nabla \tilde{f} \cdot \nabla \tilde{\zeta} \right| \leq \frac{1}{2} |p_{T_x M} - p_T|^2 \sup |\nabla \tilde{\zeta}|,$$

then

$$\begin{aligned} &\left| \sigma^{-n} \int_{M \cap F} \nabla \tilde{f} \cdot \nabla \tilde{\zeta} d\mu \right| \\ &\leq \frac{\sigma^{-n} \sup |\nabla \tilde{\zeta}|}{2} \int_{M \cap F \cap B_{\frac{\sigma}{3}}(0)} |p_{T_x M} - p_T|^2 d\mu + C \left(E_0 + \delta^{\frac{1}{2}} E_0^{\frac{1}{2}} \right) \sup |\nabla \tilde{\zeta}| \\ &\leq C \left(E_0 + \delta^{\frac{1}{2}} E_0^{\frac{1}{2}} \right) \sup |\nabla \tilde{\zeta}|, \end{aligned}$$

and if we take $\zeta = f^j$, then

$$\left| \left| \nabla^M \tilde{f} \right|^2 - \left| \nabla \tilde{f} \right|^2 \right| \leq \frac{1}{2} |p_{T_x M} - p_T|^2 \sup |\nabla \tilde{f}| \leq \frac{1}{2} |p_{T_x M} - p_T|^2,$$

so

$$\begin{aligned} &\sigma^{-n} \int_{M \cap F \cap B_{\frac{\sigma}{3}}(0)} \left| \nabla \tilde{f}^j \right|^2 d\mu \\ &\leq \sigma^{-n} \int_{M \cap F \cap B_{\frac{\sigma}{3}}(0)} \left| \nabla^M \tilde{f}^j \right|^2 d\mu + \sigma^{-n} \int_{M \cap F \cap B_{\frac{\sigma}{3}}(0)} \left| \left| \nabla^M \tilde{f}^j \right|^2 - \left| \nabla \tilde{f}^j \right|^2 \right| d\mu \\ &\leq 2E_0 + \frac{1}{2} E_0 \leq CE_0. \end{aligned}$$

By the area formula,

$$\begin{aligned}
\left| \sigma^{-n} \int_{B_{\frac{\sigma}{4}}^n(0)} \nabla f^j \cdot \nabla \zeta \theta \circ G J_G d\mathcal{L}^n \right| &= \left| \sigma^{-n} \int_{\left(B_{\frac{\sigma}{4}}^n(0) \times \mathbb{R}^k\right) \cap F} \nabla f^j \cdot \nabla \zeta \theta d\mathcal{H}^n \right| \\
&= \left| \sigma^{-n} \int_{\left(B_{\frac{\sigma}{4}}^n(0) \times \mathbb{R}^k\right) \cap F} \nabla f^j \cdot \nabla \zeta d\mu \right| \\
&\leq C \left(E_0 + \delta^{\frac{1}{2}} E_0^{\frac{1}{2}} \right) \sup |\nabla \zeta| \\
&\leq C \delta^{\frac{1}{2n+2}} E_0^{\frac{1}{2}} \sup |\nabla \zeta|,
\end{aligned}$$

$$\begin{aligned}
\sigma^{-n} \int_{B_{\frac{\sigma}{4}}^n(0)} |\nabla f^j|^2 \theta \circ G J_G d\mathcal{H}^n &= \sigma^{-n} \int_{\left(B_{\frac{\sigma}{4}}^n(0) \times \mathbb{R}^k\right) \cap F} |\nabla f^j|^2 d\mu \\
&\leq C E_0,
\end{aligned}$$

where $G : \mathbb{R}^n \rightarrow \mathbb{R}^{n+k}$ is the graph map defined by $x \in B_{\frac{\sigma}{3}}^n(0) \mapsto G(x) = (x, f(x)) \in F \subset \mathbb{R}^{n+k}$, and J_G is the Jacobian of G . Then in $B_{\sigma}^n(0)$ we have $1 \leq J_G \leq 1 + C |\nabla f|^2$, $1 \leq \theta \leq 1 + C\delta$, so

$$\begin{aligned}
\left| \sigma^{-n} \int_{B_{\sigma}^n(0)} \nabla f^j \cdot \nabla \zeta d\mathcal{L}^n \right| &\leq C \delta^{\frac{1}{2n+2}} E_0^{\frac{1}{2}} \sup |\nabla \zeta|, \\
\sigma^{-n} \int_{B_{\frac{\sigma}{4}}^n(0)} |\nabla f^j|^2 d\mathcal{L}^n &\leq C E_0.
\end{aligned}$$

Let $g^j = (C E_0)^{-\frac{1}{2}} f^j$, then we have

$$\begin{aligned}
\left| \sigma^{-n} \int_{B_{\sigma}^n(0)} \nabla g^j \cdot \nabla \zeta d\mathcal{L}^n \right| &\leq C \delta^{\frac{1}{2n+2}} \sup |\nabla \zeta|, \\
\sigma^{-n} \int_{B_{\frac{\sigma}{4}}^n(0)} |g^j|^2 d\mathcal{L}^n &\leq 1,
\end{aligned}$$

then using the Harmonic Approximation Lemma, there exist harmonic functions $v^1, \dots, v^n$ on $B_{\frac{\delta}{4}}(0)$ such that

$$\sigma^{-n} \int_{B_{\frac{\sigma}{4}}^n(0)} |Dv|^2 d\mathcal{L}^n \leq 1, \quad \sigma^{-n-2} \int_{B_{\frac{\sigma}{4}}^n(0)} |g - v|^2 d\mathcal{L}^n \leq \epsilon.$$

Let $u = (C E_0)^{\frac{1}{2}} g$, then

$$\sigma^{-n} \int_{B_{\frac{\sigma}{4}}^n(0)} |Du|^2 d\mathcal{L}^n \leq C E_0, \quad \sigma^{-n-2} \int_{B_{\frac{\sigma}{4}}^n(0)} |f - u|^2 d\mathcal{L}^n \leq C E_0 \epsilon.$$

By the interior estimate of harmonic functions,

$$\begin{aligned}
\frac{1}{\sigma} |u(0)| &\leq \frac{C\sigma^{-\frac{n}{2}}}{\sigma} \|u\|_{L^2(B_{\frac{\sigma}{4}}^n(0))} \leq \frac{C\sigma^{-\frac{n}{2}}}{\sigma} \left(\|u - f\|_{L^2(B_{\frac{\sigma}{4}}^n(0))} + \|f\|_{L^2(B_{\frac{\sigma}{4}}^n(0))} \right) \\
&\leq \frac{C}{\sigma} \left(E_0^{\frac{1}{2}} \epsilon^{\frac{1}{2}} \sigma + \delta^{\frac{1}{2n+2}} \sigma \right) \\
&\leq C \left(E_0^{\frac{1}{2}} \epsilon^{\frac{1}{2}} + \delta^{\frac{1}{2n+2}} \right) \\
&\leq C \delta^{\frac{1}{2n+2}}.
\end{aligned}$$

$$|\nabla u(0)| \leq C\sigma^{-\frac{n}{2}} \|\nabla u\|_{L^2(B_{\frac{\sigma}{4}}^n(0))} \leq CE_0^{\frac{1}{2}}.$$

Now we define the linear function $\lambda^j(x) = u^j(0) + x \cdot \nabla u^j(0)$ for $j = 1, \dots, k$, then

$$\begin{aligned}
(\eta\sigma)^{-n-2} \int_{B_{\eta\sigma}^n(0)} |f - \lambda|^2 d\mathcal{L}^n &\leq 2(\eta\sigma)^{-n-2} \int_{B_{\eta\sigma}(0)} (|f - u|^2 + |u - \lambda|^2) d\mathcal{L}^n \\
&\leq 2\eta^{-n-2} E_0 \epsilon + 2\omega_n (\eta\sigma)^{-2} \left(\sup_{B_{\frac{\sigma}{4}}^n(0)} |u - \lambda| \right)^2 \\
&\leq C\eta^{-n-2} E_0 \epsilon + C\eta^2 E_0,
\end{aligned}$$

where we use $\sup_{B_{\frac{\sigma}{4}}^n(0)} |u - \lambda| \leq C\eta^2 \sigma^{1-\frac{n}{2}} \|\nabla u\|_{L^2(B_{\frac{\sigma}{4}}^n(0))} \leq C\eta^2 \sigma E_0^{\frac{1}{2}}$. Now let S be the n -dimensional subspace graph $(\lambda - \lambda(0))$, and let $\tau = (0, \lambda(0))$. Observing that $\text{dist}(x, \tau + S) \leq |f(x') - \lambda(x')|$ for any $x = (x', f(x')) \in B_{\eta\sigma}(0) \cap F$, we have

$$(\eta\sigma)^{-n-2} \int_{B_{\eta\sigma}(\tau) \cap F} \text{dist}(x - \tau, S)^2 d\mathcal{L}^n \leq C\eta^{-n-2} \epsilon E_0 + C\eta^2 E_0.$$

So we have

$$\begin{aligned}
&(\eta\sigma)^{-n-2} \int_{B_{\eta\sigma}(\tau)} \text{dist}(x - \tau, S)^2 d\mu \\
&= (\eta\sigma)^{-n-2} \int_{B_{\eta\sigma}(\tau) \cap F} \text{dist}(x - \tau, S)^2 d\mu \\
&\quad + (\eta\sigma)^{-n-2} \int_{B_{\eta\sigma}(\tau) - F} \text{dist}(x - \tau, S)^2 d\mu \\
&\leq C\eta^{-n-2} \epsilon E_0 + C\eta^2 E_0 + (\eta\sigma)^{-n-2} \cdot C\delta^{\frac{1}{n+1}} \sigma^2 \cdot \mu(B_{\eta\sigma}(\tau) - F) \\
&\leq C\eta^{-n-2} \epsilon E_0 + C\eta^2 E_0 + C\eta^{-n-2} \delta^{\frac{1}{n+1}} \sigma^{-n} \cdot CE_0 \sigma^n \\
&\leq C\eta^{-n-2} \left(\epsilon + \delta^{\frac{1}{n+1}} \right) E_0 + C\eta^2 E_0.
\end{aligned}$$

Then we have

$$\begin{aligned} E(\tau, \frac{\eta\sigma}{2}, S) &\leq C\sigma^{-n} \int_{B_{\eta\sigma}(\tau)} \left(\frac{\text{dist}(x, \tau + S)}{\rho} \right)^2 + C \left(\sigma^{p-n} \int_{B_{\eta\sigma}(\tau)} |\underline{H}|^p d\mu \right)^{\frac{2}{p}} \\ &\leq C\eta^{-n-2}(\epsilon + \delta^{\frac{1}{n+1}})E_0 + C(\eta^2 + \delta)E_0. \end{aligned}$$

Because $|\tau| \leq C\delta^{\frac{1}{2n+2}}\sigma$, so $C\delta^{\frac{1}{2n+2}} < \frac{\eta}{4} \implies B_{\frac{\eta\sigma}{4}}(0) \subset B_{\frac{\eta\sigma}{2}}(\tau)$, then clearly we have

$$E(0, \frac{\eta\sigma}{4}, S) \leq C\eta^{-n-2} \left(\epsilon + \delta^{\frac{1}{n+1}} \right) E_0 + C\eta^2 E_0.$$

Finally we can prove this lemma. Select $\eta = \eta(n, k, p)$ so that $C\eta^2 \leq \frac{1}{4}(\frac{\eta}{4})^{2(1-\frac{n}{p})}$, and choose $\epsilon = \epsilon(n, k, p)$ so that $C\eta^{-n-2}\epsilon \leq \frac{1}{4}(\frac{\eta}{4})^{2(1-\frac{n}{p})}$, and finally choose $\delta \leq \delta_0(n, k, p)$ with δ_0 sufficiently small so that $B_{\frac{\eta\sigma}{4}}(0) \subset B_{\frac{\eta\sigma}{2}}(\tau)$, and so that the harmonic approximation lemma is valid provided ϵ is small enough, and so that $C\eta^{-n-2}\delta^{\frac{1}{n+1}} \leq \frac{1}{4}(\frac{\eta}{4})^{2(1-\frac{n}{p})}$. Then we obtain

$$E(0, \tilde{\eta}\sigma, S) \leq \tilde{\eta}^{2(1-\frac{n}{p})} E_0, \quad \tilde{\eta} = \frac{\eta}{4}.$$

Since

$$((\tilde{\eta}\sigma)^{p-n} \int_{B_{\tilde{\eta}\sigma}(0)} |\underline{H}|^p d\mu)^{\frac{1}{p}} \leq \tilde{\eta}^{1-\frac{n}{p}} \left(\sigma^{p-n} \int_{B_\sigma(0)} |\underline{H}|^p d\mu \right)^{\frac{1}{p}},$$

we conclude that

$$E_*(0, \tilde{\eta}\sigma, S) \leq \tilde{\eta}^{2(1-\frac{n}{p})} E_*(0, \sigma, T).$$

□

3.5 Proof of Allard Regularity Theorem

Assume that $V = v(M, \theta)$ is a rectifiable varifold with generalized mean curvature $\underline{H}$ in the open set $U \subset \mathbb{R}^{n+k}$, satisfying the following conditions:

$$\begin{cases} \theta \geq 1 \text{ } \mu\text{-a.e.}, \\ 0 \in \text{spt}\mu, \\ B_\rho(0) \subset U, \\ \frac{1}{\omega_n \rho^n} \mu(B_\rho(0)) \leq 1 + \delta, \\ \left(\rho^{p-n} \int_{B_\rho(0)} |\underline{H}|^p d\mu \right)^{\frac{1}{p}} \leq \delta, \end{cases}$$

then we have the following theorem:

Theorem 3.9 (Allard Regularity Theorem). *If $p > n$ is arbitrary, then there are constants $\gamma = \gamma(n, k, p)$, $\delta_0 = \delta_0(n, k, p) \in (0, \frac{1}{16}]$ such that if $\delta \in (0, \delta_0]$ and*

if the conditions above hold, then there exists an orthogonal transformation Q of $\mathbb{R}^{n+k}$ and a $u = (u^1, \dots, u^k) \in C^{1,1-\frac{n}{p}}(B_{\gamma\rho}^n(0); \mathbb{R}^k)$ with

$$Du(0) = 0, \text{spt } u \cap B_{\gamma\rho}(0) = Q(\text{graph } u) \cap B_{\gamma\rho}(0),$$

and

$$\rho^{-1} \sup |u| + \sup |Du| + \rho^{1-\frac{n}{p}} \sup_{x,y \in B_{\gamma\rho}^n(0), x \neq y} |x-y|^{-(1-\frac{n}{p})} |Du(x) - Du(y)| \leq C\delta^{\frac{1}{2n+2}},$$

where $C = C(n, k, p) > 0$ and $\gamma = \gamma(n, k, p) \in (0, 1)$.

Lemma 3.10. *There exists a constant $\delta_0 = \delta_0(n, k, p)$ such that if $\delta \in (0, \delta_0]$, if the conditions above hold, and if $T = T(0, 4\delta\rho)$, then for each $\theta \in (0, 1)$,*

$$\frac{1}{\rho^n} \int_{B_{\theta\rho}(0)} |p_{T_x M} - p_{T_0}|^2 d\mu_V \leq C\delta^{\frac{1}{n+1}},$$

where $C = C(\theta, n, k, p)$.

Proof. Let $t \in [\delta, 1)$, using the monotonicity formula with $\sigma = \delta t$, $h(x) = \left| p_{T_0^\perp} \left(\frac{x}{|x|} \right) \right|^2$,

$$\begin{aligned} \int_{\partial B_t(0)} h |\nabla^M r| d\mu_V &\leq \delta^{1-n} \int_{\partial B_{\delta t}(0)} h |\nabla^M r| d\mu_V + n \int_{B_1(0)} r^{-n} |D^\perp r|^2 d\mu_V \\ &\quad + C \int_{B_1(0) - B_{\delta^2}(0)} (r^{-n} |D^\perp r| + r^{1-n} h |\underline{H}|) d\mu_V. \end{aligned}$$

Here using $|D^\perp r| \leq \delta^{\frac{1}{2}} + \delta^{-\frac{1}{2}} |D^\perp r|^2$ and $\int_{B_1(0) - B_{\delta^2}(0)} r^{-n} \leq C |\log \delta|$, we obtain

$$\begin{aligned} \int_{\partial B_t(0)} h |\nabla^M r| d\mu_V &\leq \delta^{1-n} \int_{\partial B_{\delta t}(0)} h |\nabla^M r| d\mu_V + C\delta^{-\frac{1}{2}} \int_{B_1(0)} r^{-n} |D^\perp r|^2 d\mu_V \\ &\quad + C\delta^{\frac{1}{2}} |\log \delta| + \int_{B_1(0)} r^{1-n} h |\underline{H}| d\mu_V. \end{aligned}$$

Also we have

$$\int_{B_1} r^{1-n} h |\underline{H}| d\mu_V \leq C \|\underline{H}\|_{L^p(\mu_V \llcorner B_1(0))} \leq C\delta,$$

so

$$\int_{\partial B_t(0)} h |\nabla^M r| d\mu_V \leq \delta^{1-n} \int_{\partial B_{\delta t}(0)} h |\nabla^M r| d\mu_V + C\delta |\log \delta|.$$

Integrating over $t \in (\delta, 1)$ and using the Co-area Formula, we obtain

$$\int_{B_1(0) - B_\delta(0)} h |\nabla^M r|^2 d\mu_V \leq \delta^{-n} \int_{B_\delta(0) - B_{\delta^2}(0)} h |\nabla^M r|^2 d\mu_V + C\delta |\log \delta|,$$

so

$$\int_{B_1(0)} h |\nabla^M r|^2 d\mu_V \leq (1 + \delta^{-n}) \int_{B_\delta(0)} h |\nabla^M r|^2 d\mu_V + C\delta^{\frac{1}{2}} |\log \delta|.$$

Since $|\nabla^M r|^2 = 1 - |D^\perp r|^2$,

$$\begin{aligned} \int_{B_1(0)} h d\mu_V &\leq (1 + \delta^{-n}) \int_{B_\delta(0)} h (1 - |D^\perp r|^2) d\mu_V + \int_{B_1(0)} h |D^\perp r|^2 d\mu_V + C\delta^{\frac{1}{2}} |\log \delta| \\ &\leq (1 + \delta^{-n}) \int_{B_\delta(0)} h d\mu_V + \int_{B_1(0) - B_\delta(0)} h |D^\perp r|^2 d\mu_V + C\delta^{\frac{1}{2}} |\log \delta| \\ &\leq (1 + \delta^{-n}) \int_{B_\delta(0)} h d\mu_V + C \int_{B_1(0) - B_\delta(0)} r^{-n} |D^\perp r|^2 d\mu_V + C\delta^{\frac{1}{2}} |\log \delta| \\ &\leq (1 + \delta^{-n}) \int_{B_\delta(0)} h d\mu_V + C\delta + C\delta^{\frac{1}{2}} |\log \delta|. \end{aligned}$$

By $h(x) = \left| p_{T_0} \left(\frac{x}{|x|} \right) \right|^2 \leq C\delta^{\frac{1}{n+1}}$ by the Affine Approximation Lemma,

$$\begin{aligned} \int_{B_1(0)} \left| p_{T_0^\perp}(x) \right|^2 d\mu_V &\leq \int_{B_1(0)} h d\mu_V \\ &\leq C(1 + \delta^{-n}) \int_{B_\delta(0)} \delta^{\frac{1}{n+1}} + C\delta + C\delta^{\frac{1}{2}} |\log \delta| \\ &\leq C\delta^{\frac{1}{n+1}}. \end{aligned}$$

□

Eventually we can prove the main theorem - Allard Regularity Theorem.

Proof of Allard Regularity Theorem. Take $\xi \in B_{\frac{\delta\rho}{2}}(0) \cap \text{spt} V$ and $\sigma \in (0, \frac{\delta\rho}{2}]$, and let S_0 be an arbitrary n -dimensional subspace of $\mathbb{R}^{n+k}$. By the Tilt-Excess Decay Theorem, there exist constants $\delta_0 = \delta_0(n, k, p)$, $\eta = \eta(n, k, p)$ such that if $\delta \leq \delta_0$, then

$$E_*(\xi, \eta\sigma, S_1) \leq \eta^{2(1-\frac{n}{p})} E_*(\xi, \sigma, S_0)$$

for suitable S_1 . Repeat this process, we can prove that if $\xi \in \text{spt} V \cap B_{\frac{\delta\rho}{2}}(0)$, and let $\sigma_0 = \frac{\delta\rho}{2}$, then there exists a sequence $\{S_n\}_{n=1}^\infty$ of n -dimensional subspace of $\mathbb{R}^{n+k}$ such that

$$E_*(\xi, \eta^j \sigma_0, S_j) \leq \eta^{2(1-\frac{n}{p})} E_*(\xi, \frac{\eta^{j-1} \rho}{2}, S_{j-1}) \leq \eta^{2(1-\frac{n}{p})j} E_*(\xi, \sigma_0, S_0)$$

for each $j \geq 1$.

Let $T_0 = T(0, 2\sigma_0)$, then we have $E(0, \sigma_0, T_0) \leq C\delta^{\frac{1}{n+1}}$, and hence $E(\xi, \frac{\sigma_0}{2}, T_0) \leq 2^n C\delta^{\frac{1}{n+1}}$ for each $\xi \in \text{spt} V \cap B_{\frac{\sigma_0}{2}}(0)$. We always take $S_0 = T_0$, then

$$E_*(\xi, \eta^j \sigma_0, S_j) \leq \eta^{2(1-\frac{n}{p})} E_*(\xi, \frac{\eta^{j-1} \sigma_0}{2}, S_{j-1}) \leq \eta^{2(1-\frac{n}{p})j} E_0,$$

where $E_0 = E_*(0, \sigma_0, T_0)$. Particularly for each $j \geq 1$ we have

$$|p_{S_j} - p_{S_{j-1}}|^2 \leq CE_*(\xi, \eta^{j-1}\sigma_0, S_{j-1}) \leq C\eta^{2(1-\frac{n}{p})j} E_*(\xi, \sigma_0, S_0).$$

Now sum from $j+1$ to l ,

$$|p_{S_l} - p_{S_j}|^2 \leq C\eta^{2(1-\frac{n}{p})j} E_*(\xi, \sigma_0, S_0)$$

for any $l \geq j \geq 0$. Therefore, $\{p_{S_i}\}$ is a Cauchy sequence, so there exists $S(\xi)$ such that

$$|p_{S(\xi)} - p_{S_j}|^2 \leq C\eta^{2(1-\frac{n}{p})j}$$

for any $j \geq 0$. In particular,

$$|p_{S(\xi)} - p_{T_0}|^2 \leq CE_0.$$

Now for arbitrary $\sigma \in [0, \frac{\sigma_0}{2}]$, we can choose $j \geq 0$ such that $\frac{\eta^j \sigma_0}{2} < \sigma \leq \frac{\eta^{j+1} \sigma_0}{2}$. Then

$$E_*(\xi, \sigma, S(\xi)) \leq C \left(\frac{\sigma}{\sigma_0} \right)^{2(1-\frac{n}{p})} E_0$$

for each $\xi \in B_{\frac{\sigma_0}{2}}(0) \cap \text{spt} V$ and each $0 < \sigma \leq \frac{\sigma_0}{2}$.

Without loss of generality, we can suppose $T_0 = \mathbb{R}^n \times \{0\}$. Then by the Lipschitz Approximation Theorem, if $L_0 \in (0, \frac{1}{4}]$ is given, and if $\delta \leq \delta_0 L_0^{2n+2}$ for suitable $\delta_0 = \delta_0(n, k, p)$, then

$$\text{spt} V \cap B_{\frac{\sigma_0}{4}}(0) = \text{graph} f \cap B_{\frac{\sigma_0}{4}}(0)$$

for some Lipschitz function $f : B_{\frac{\sigma_0}{2}}^n(0) \rightarrow \mathbb{R}^k$ with $\text{Lip} f \leq L_0$.

For such f , let $F = \text{graph} f$, $\xi = (\xi', f(\xi')) \in F$, and we have

$$\lim_{\sigma \rightarrow 0+} \sigma^{-n} \int_{B_\sigma(\xi) \cap F} |p_{T_x F} - p_{S(\xi)}|^2 d\mathcal{H}^n = 0$$

for $\mathcal{H}^n$ -a.e. $\xi \in F \cap B_{\frac{\sigma_0}{2}}(0)$. And at such a point ξ , $S(\xi)$ is the approximate tangent space of F . So

$$\sigma^{-n} \int_{B_\sigma(\xi) \cap F} |p_{T_x F} - p_{T_\xi F}|^2 d\mathcal{H}^n \leq C \left(\frac{\sigma}{\sigma_0} \right)^{2(1-\frac{n}{p})} E_0$$

for all $0 < \sigma \leq \frac{\sigma_0}{2}$.

Now, the orthogonal projection $p_{T_\xi F}$ is given by

$$p_{T_\xi F}(v) = \sum_{j=1}^n (\tau_j \cdot v) \tau_j,$$

where $\{\tau_j\}$ is an orthonormal basis for $T_\xi F$. In fact $p_{T_\xi F}$ has matrix P_ξ of the form

$$P_\xi = \begin{pmatrix} I_{n \times n} & Df(\xi') \\ (Df(\xi'))' & O_{k \times k} \end{pmatrix} + \mathcal{F}(Df(\xi'))$$

where $\mathcal{F}(p)$ is a real analytic function of $p = \{p_{ij}\}_{i=1, \dots, n; j=1, \dots, k} \in \mathbb{R}^{nk}$ with $\mathcal{F}(0) = 0$, $D_p \mathcal{F}(0) = 0$, and hence

$$|\mathcal{F}(p_1) - \mathcal{F}(p_2)| \leq C(n, k) (|p_1| + |p_2|) |p_1 - p_2|$$

for $|p_1|, |p_2| \leq 1$. Evidently when L_0 is sufficiently small, we have

$$|Df(x') - Df(\xi')|^2 \leq |p_{T_{x'} F} - p_{T_{\xi'} F}|^2 \leq 3 |Df(x') - Df(\xi')|^2.$$

So

$$\sigma^{-n} \int_{B_\sigma^n(\xi')} |Df(x) - Df(\xi)|^2 d\mathcal{L}^n \leq C \left(\frac{\sigma}{\sigma_0} \right)^{2(1-\frac{n}{p})} E_0$$

for any $0 < \sigma < \frac{\sigma_0}{4}$. For μ -a.e. $x_1, x_2 \in \text{spt} V \cap B_{\frac{\sigma_0}{8}}(0)$ we can use the formula above with $\sigma = |x_1 - x_2|$ and $\xi = x_1, x_2$. Since

$$|Df(x_1) - Df(x_2)|^2 \leq 2 |Df(x) - Df(x_1)|^2 + 2 |Df(x) - Df(x_2)|^2$$

for $x \in B_\sigma^n(x_1) \cap B_\sigma^n(x_2) \supset B_{\frac{\sigma}{2}}^n\left(\frac{x_1+x_2}{2}\right)$. Then we conclude

$$|Df(x_1) - Df(x_2)| \leq C \left(\frac{|x_1 - x_2|}{\sigma_0} \right)^{1-\frac{n}{p}} E_0^{\frac{1}{2}}$$

for $\mathcal{L}^n$ -a.e. $x_1, x_2 \in B_{\frac{\sigma_0}{4}}^n(0)$. Then choosing $\delta = \delta(n, k, p)$ to satisfy the smallness condition, the theorem is now proved with $u = f$ and $\gamma = \frac{\delta}{4}$. $\square$